Project Proposal (Proposed Solution) Report

Date	20 October 2026
Team ID	Individual project (no team)
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	5 Marks

The proposal describes an AI-powered web application that turns a traveller's preferences into a personalised, day-by-day itinerary, saves it, and makes trip planning faster and less stressful.

Project Overview

Objective	Build and deploy a full-stack generative AI application that creates personalised travel itineraries from destination, number of days, budget, vibe and interests.
Scope	Preference form, destination suggestions, AI itinerary generation with timings, travel times and costs, saving, viewing and deleting trips, copy-as-text export, and cloud deployment.

Problem Statement

Description	Travellers spend hours researching across many sites and still end up with generic plans that ignore their budget, pace and mood.
Impact	Solving this saves planning time, reduces decision fatigue, improves budget awareness, and gives travellers a clear plan they can store and reuse.

Proposed Solution

Approach	A React frontend sends the preferences to a FastAPI backend. The backend builds a structured prompt (group nearby places, order by time, add travel time and cost, match the vibe), calls the Groq LLM in JSON mode, and returns a structured itinerary. Trips are stored in Supabase.
Key Features	- Preference form with vibe-based personalisation - Time-ordered day-wise itinerary with travel times and cost estimates - Destination suggestions - Save, view and delete trips (Supabase) - Copy itinerary as text - Cloud deployment on Vercel and Render

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware: Computing resources	Development machine; cloud hosting	Personal laptop for development; Render and Vercel free tiers for hosting
Hardware: Storage	Database storage	Supabase free tier (managed PostgreSQL)
Software: Frameworks	Web frameworks	FastAPI (backend), React 18 + Vite (frontend)
Software: Libraries	Additional libraries	groq, supabase-py, python-dotenv, pydantic, uvicorn
Software: Development environment	IDE and tools	VS Code, Git, GitHub, Postman / FastAPI docs page
Data	Source, size, format	No training dataset; itineraries are generated by the LLM at request time; trips stored as JSON rows in Supabase